Good and Bad Relations

Topics List

- Good Relations versus Bad Relations
- What’s so Bad about a Bad Relation?
- How do we get a Good Relation?

Good Relations versus Bad Relations

- A relation is *good* if it records data about one object of interest. A relation that is not good is *bad*.
- In relational database design, we can attempt to build good relations. And, we can convert bad relations into good relations.
- What is a relation “about”? The key to the answer is usually the *Primary key*.

Good Relation Examples

- *The Branch* relation is good because the key is branchNo, which identifies different branches, and each attribute (street, city, state, zipCode, and telNo) is a piece of information about each Branch.
- From the relation, we can see that for each branch there is one branchNo value, one street value, one city value, one state value, one zipCode value, and one telNo value.

Branch

branchNo	street	city	state	zipCode	telNo
B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
B002	City Center Plaza	Seattle	WA	98122	206-555-6756
B003	14 – 8th Avenue	New York	NY	10012	212-371-3000
B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131

Primary key branchNo

Good Relation Examples

- Likewise, the *Staff* relation is good because the key is StaffNo, which identifies different Staff members, and each attribute (firstName, lastName, position, salary, and branchNo) is a piece of information about each Staff member.
- We can see that for each staff member there is one staffNo value, one firstName value, one lastName value, one position value, one salary value, and one branchNo value.
- The branchNo identifies the branch where each staff members works, and it is a branch from the Branch relation.

Staff

staffNo	firstName	lastName	position	salary	branchNo
S1500	Tom	Daniels	Manager	46000	B001
S0003	Sally	Adams	Assistant	30000	B001
S0010	Mary	Martinez	Manager	50000	B002
S3250	Robert	Chin	Supervisor	32000	B002
S2250	Sally	Stern	Manager	48000	B004
S0415	Art	Peters	Manager	41000	B003

Primary key staffNo
Foreign key branchNo references Branch(branchNo)

Bad Relation Examples

StaffBranch

staffNo	firstName	lastName	position	salary	branchNo	street	city	state	zipCode	telNo
S1500	Tom	Daniels	Manager	46000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0003	Sally	Adams	Assistant	30000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0010	Mary	Martinez	Manager	50000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S3250	Robert	Chin	Supervisor	32000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S2250	Sally	Stern	Manager	48000	B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131
S0415	Art	Peters	Manager	41000	B003	14 – 8th Avenue	New York	NY	10012	212-371-3000

Primary key staffNo

- The key for this relation is staffNo, which infers that the relation is about staff members – which it is.
- But the relation also contains attributes – branchAddress (street, city, state, zipCode) and telNo – that are not about Staff but about the branch where the staff is located!
- So, this relation is *bad* and it needs to be converted into 2 relations as in the previous 2 slides.

Bad Relation Examples

Work

employeeId	firstName	lastName	stateCode	state	jobCode	job
E001	Alice	Smith	26	Michigan	J01	Chef
E001	Alice	Smith	26	Michigan	J02	Waiter
E002	Bob	Kennedy	66	Wyoming	J02	Waiter
E002	Bob	Kennedy	66	Wyoming	J03	Bartender
E003	Caroline	McDaid	26	Michigan	J01	Chef

Primary key employeeId, jobCode

- The key for this relation is employeeId, jobCode, which infers that the relation is about employees and the work they do – which it is.
- But the relation also contains attributes – firstName, lastName, stateCode, and state, which are about employees only and each time an employee is assigned a job, those values are replicated!
- Likewise, the relation also contains an attribute – job, which is about job assigned only and each time a job is assigned to an employee, this value is replicated also!
- So, this relation is *bad* and it needs to be transformed.

Bad Relation Examples

Student					
studentId	**firstName**	**lastName**	**emailAddress**	**module**	**GPA**
100111	John	Doe	john.doe@usna.edu	NN204, S1204, IT221	2, 3, 3
192244	Matt	Smith	matt.smith@usna.edu	SM223, EE301	4, 4
113221	Melinda	Black	melinda.black@usna.edu	S1204	3
390112	Tom	Johnson	tom.johnson@usna.edu	NN204, S1204, IT221	4, 2, 3
Primary key studentId					

- The key for this relation is StudentId, which infers that the relation is about Students – which it is.
- But the relation does not conform to one of the rules for relations (Each cell of relation contains exactly one atomic (single) value) as the module and GPA columns hold multiple values instead of one single value.
- So, this relation is *bad* and it needs to be transformed.
- Note: If we design our relations properly, they are normally pretty *good*.

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What's so Bad about a Bad Relation?

StaffBranch

staffNo	firstName	lastName	position	salary	branchNo	street	city	state	zipCode	telNo
S1500	Tom	Daniels	Manager	46000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0003	Sally	Adams	Assistant	30000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0010	Mary	Martinez	Manager	50000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S3250	Robert	Chin	Supervisor	32000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S2250	Sally	Stern	Manager	48000	B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131
S0415	Art	Peters	Manager	41000	B003	14 – 8th Avenue	New York	NY	10012	212-371-3000

Primary key staffNo

- We have already established that this relation is bad. So, what is so bad about a bad relation?
- We will examine the reasons under the following three anomalies:
  - Insert
  - Update
  - Delete

What's so Bad about a Bad Relation?

Insert Anomaly:

- We want to add a new branch B005 in New York but for now there are no employees working at the branch.
- Since the primary key for the relation is staffNo, we cannot add a record until we have an employee working at the new branch (Entity Integrity)!

Not permitted →

StaffBranch

staffNo	firstName	lastName	position	salary	branchNo	street	city	state	zipCode	telNo
S1500	Tom	Daniels	Manager	46000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0003	Sally	Adams	Assistant	30000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0010	Mary	Martinez	Manager	50000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S3250	Robert	Chin	Supervisor	32000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S2250	Sally	Stern	Manager	48000	B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131
S0415	Art	Peters	Manager	41000	B003	14 – 8th Avenue	New York	NY	10012	212-371-3000
null	null	null	null	null	B005	21 - 17th Avenue	New York	NY	10078	212-349-9988

Primary key staffNo

What's so Bad about a Bad Relation?

Update Anomaly:

- What happens if branch B002 changes its telNo from 206-555-6756 to 206-555-7023?

StaffBranch

staffNo	firstName	lastName	position	salary	branchNo	street	city	state	zipCode	telNo
S1500	Tom	Daniels	Manager	46000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0003	Sally	Adams	Assistant	30000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0010	Mary	Martinez	Manager	50000	B002	City Center Plaza	Seattle	WA	98122	206-555-7023
S3250	Robert	Chin	Supervisor	32000	B002	City Center Plaza	Seattle	WA	98122	206-555-7023
S2250	Sally	Stern	Manager	48000	B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131
S0415	Art	Peters	Manager	41000	B003	14 – 8th Avenue	New York	NY	10012	212-371-3000

Primary key staffNo

Multiple changes

- We must search every record/row and change every instance of 206-555-6756 to 206-555-7023 in records involving branch B002.

What's so Bad about a Bad Relation?

Update Anomaly:

- This one change in the real world may make for many changes in the database.
- What if we mess up and end up not making all changes?

StaffBranch

staffNo	firstName	lastName	position	salary	branchNo	street	city	state	zipCode	telNo
S1500	Tom	Daniels	Manager	46000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0003	Sally	Adams	Assistant	30000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0010	Mary	Martinez	Manager	50000	B002	City Center Plaza	Seattle	WA	98122	206-555-7023
S3250	Robert	Chin	Supervisor	32000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S2250	Sally	Stern	Manager	48000	B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131
S0415	Art	Peters	Manager	41000	B003	14 – 8th Avenue	New York	NY	10012	212-371-3000

Primary key staffNo

Two different telNo values

- Now, what is the telephone number for branch B002?

What’s so Bad about a Bad Relation?

Delete Anomaly:

- What if staff member Sally Stern leaves the company and her record is deleted?

StaffBranch

staffNo	firstName	lastName	position	salary	branchNo	street	city	state	zipCode	telNo
S1500	Tom	Daniels	Manager	46000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0003	Sally	Adams	Assistant	30000	B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
S0010	Mary	Martinez	Manager	50000	B002	City Center Plaza	Seattle	WA	98122	206-555-7023
S3250	Robert	Chin	Supervisor	32000	B002	City Center Plaza	Seattle	WA	98122	206-555-6756
S0415	Art	Peters	Manager	41000	B003	14 – 8th Avenue	New York	NY	10012	212-371-3000

Primary key staffNo

- A consequence is that we lose all information about branch B004 also.

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How do we get a Good Relation?

- As previously stated in relational database design, we can attempt to build good relations and that is where we will go from here…
- We also previously stated that we can convert bad relations into good relations and we will look at this later.